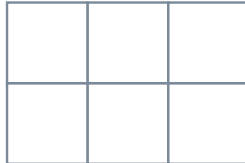


Practice Set No. 3

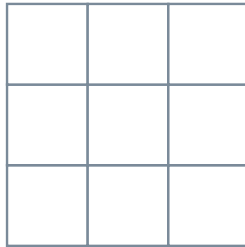
Dominoes and the Checkerboard

A domino is a tile that covers two squares side by side. Take your time — the trick is to notice, not to compute.

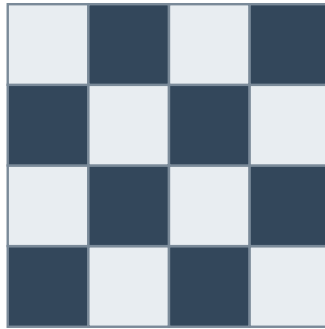
1. On the last sheet you will find paper dominoes and a small board to cut out. Each **domino** covers exactly two squares that sit next to each other. Cover the board completely — no gaps, no overlaps, nothing hanging off the edge. How many dominoes did it take?



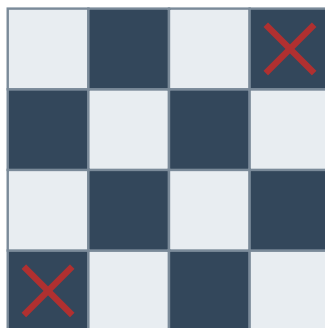
2. Here is a board with nine squares. Try to cover it with dominoes the same way. Can you do it? Explain why or why not.



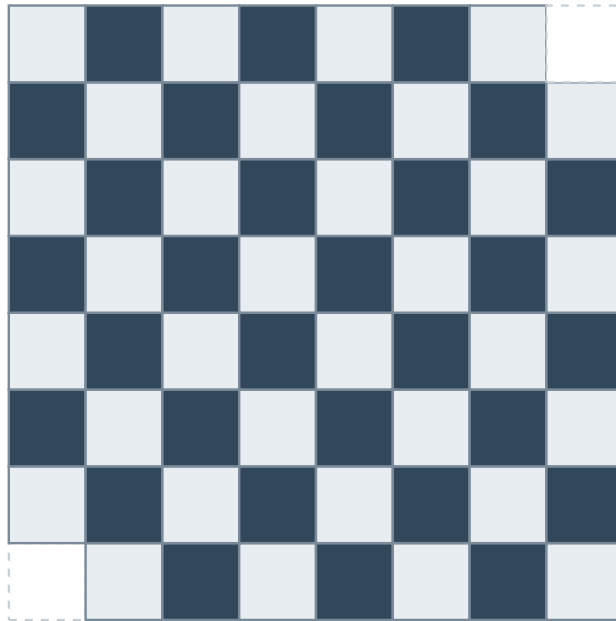
3. This board is shaded like a checkerboard. Picture covering it with dominoes — each one lies flat across two squares that share an edge. Pick any spot for a domino: **what two colors does it land on?** Is that always true, no matter where it goes?



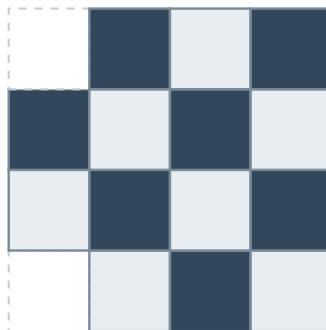
4. On this board the two squares marked **X** have been taken away — they sit in opposite corners, and notice they are the **same color**. Now try to cover the 14 squares that remain. Can you? Count how many dark and how many light squares are left, and use problem 3 to explain.



5. A full chessboard has $8 \times 8 = 64$ squares. Two opposite corners are cut off, leaving 62 squares. Could exactly **31 dominoes** cover what is left? Explain your answer using what you found in problem 4.



6. This time two squares are removed — but **one is dark and one is light**, so the colors are balanced again. Does the covering work now? Try it on the board below, and say what the checkerboard coloring was really checking all along.



Answer Key — for parents

1. **3 dominoes.** The board has 6 squares and each domino covers 2, so $6 \div 2 = 3$. (Any full covering must split the squares into pairs.)
2. **No.** Each domino covers 2 squares, so any board that can be covered must have an **even** number of squares. Nine is odd, so one square is always left stranded. (*This is the odd/even tool from Saturday Test No. 7.*)
3. Every domino lands on **exactly one dark and one light** square. It always covers two neighbors, and on a checkerboard neighbors are always different colors — so this is true no matter where the domino goes.
4. **No.** The two opposite corners are the same color (both dark), so removing them leaves **6 dark and 8 light** squares. But every domino needs one of each color — 6 dominoes use up the 6 dark squares, and 2 light squares are left with no partner. Impossible.
5. **No.** Opposite corners of a chessboard share a color, so cutting them off leaves **30 of one color and 32 of the other**. Each of the 31 dominoes covers one of each color, which would need 31 and 31. The 2 extra squares of the majority color can never be covered. (*This is the classic “mutilated chessboard.”*)
6. **Yes** — with one dark and one light square gone, the colors are balanced (7 and 7 on the 4×4), and the board can be covered. The coloring was never really about corners; it was checking whether **the dark and light counts stay equal**. If they don't, no covering can exist. (*In fact, on any chessboard, removing one dark and one light square always leaves something dominoes can cover.*)